

Bonus Assignment#1: Grid Generation over NACA 63-412 Airfoil

Instructions

1. This assignment includes 10 bonus marks, which will be added on top of your scores from course assignments, quizzes, and examinations. It consists of three parts:
 - (a) Spline fitting of the NACA 63-412 airfoil data. The coordinate data for the NACA 63-412 airfoil is also uploaded in moodle along with this file.
 - (b) Generate a 40×40 grid around NACA 63-412 airfoil using trans-finite interpolation (TFI) technique.
 - (c) Smooth the grid obtained from TFI using the elliptic grid generation method.
2. You will have to plan everything of your own without the help of the course instructor and TAs at any point of time. You may however ask for clarification of any doubt about the problem statement.
3. You must to submit a report giving every finer detail of your (a) formulation, (b) numerical implementation, and (c) computer code along with README.txt file explaining the commands to run your program and produce the plot. Any guess work (missing information) will invalidate your submission.
4. The assignment will not be evaluated even if the computer code runs and generates correct results if the report is incomplete (full description as mentioned above is not provided therein).
5. You will have to appear for a viva.

Problem Statement

Generate a 40×40 grid around NACA 63-412 airfoil using (a) transfinite interpolation (TFI) technique and (b) Elliptic grid generation. The physical and computational domain is shown in the Figure 1. You must use spline to represent the airfoil contour.

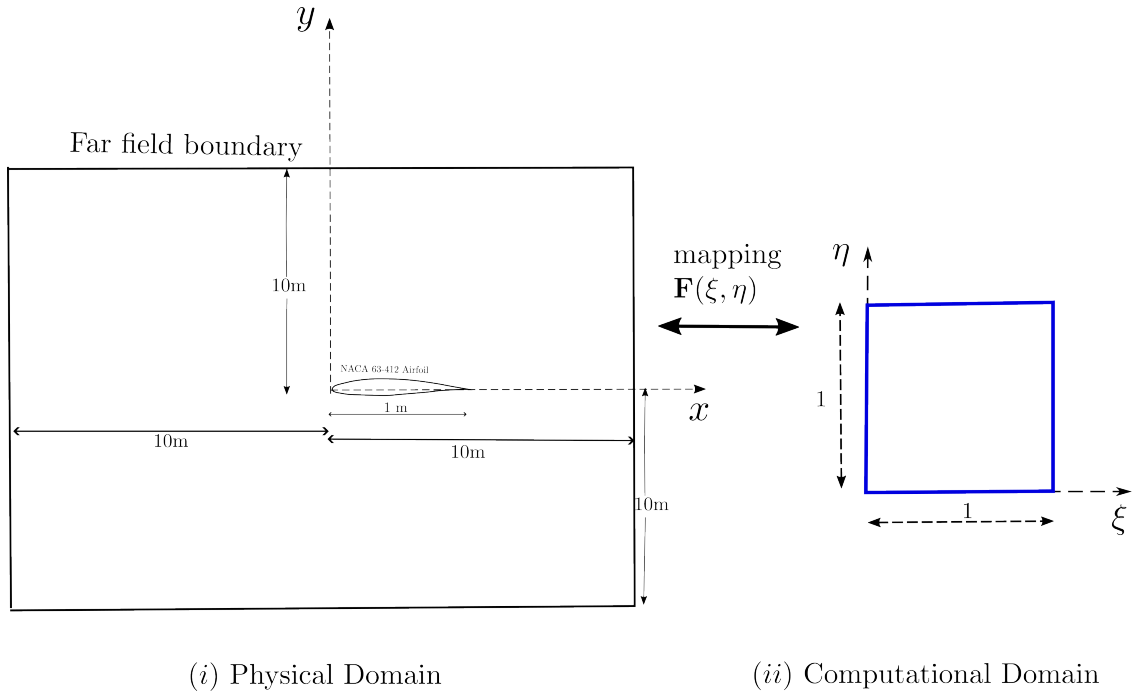


Figure 1: NACA 63-412 Airfoil

General Instructions

- **Checklist for submission:**

1. Submit the source codes for
 - Spline fitting of the airfoil
 - TFI grid generation
 - Elliptic grid generations

The codes should be written in Python, C, or C++.

2. Include a detailed report as specified in point number 3 at the beginning of this document.

- **Instruction for submission:**

- Rename your program file as your roll number (example: 204010006.c).
- Rename the report as your roll number (example: 204010006.pdf)
- Upload all the documents separately as stated in the Checklist above.
- Don't submit zip file.

- **Notes:**

- Marks will be given only if the program is working and showing correct result.
- Assignment will not be evaluated if “instruction for submission” are not followed properly.

NOTE: Bonus marks will be distributed among students whose submissions have more than 30% similarity detected by the MOSS plagiarism tool. To receive full marks, it is recommended that you write your own code independently, ensuring it is entirely correct.

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